

LLM for Teens

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■ Agentic rag

Regular RAG: fixed pipeline, search once, then answer.

Agentic RAG: agent decides, searches, checks, searches again if needed, then answers.

Comparison:

Steps: Regular RAG uses one retrieval. Agentic RAG uses multiple, agent-driven steps.

Sources: Regular RAG usually uses one. Agentic RAG uses many (DBs, APIs, web).

Query: Regular RAG uses it as-is. Agentic RAG rewrites or decomposes it.

Speed: Regular RAG is fast and cheap. Agentic RAG is slower and costlier.

Best for: Regular RAG suits simple lookups. Agentic RAG suits multi-step, multi-source tasks.

Example: "Compare France vs Germany sales and flag recalls."

Regular RAG does one search and gives an incomplete answer.

Agentic RAG fetches sales, compares, checks recalls, then answers.

■ Agentic rag

Traditional RAG works like a fixed recipe. You ask a question, it searches once, and the language model answers from what it found. It doesn't check if the search was good enough, and it doesn't search again.

Agentic RAG is different. An agent is in charge. It reads the question, decides which sources to search, looks at the results, and searches again if something is missing.

So the simplest way to say it: traditional RAG searches once and answers. Agentic RAG plans, searches, checks, and keeps going until it has enough to answer well.

One example. Ask both systems to compare sales in France and Germany and flag any recalls. Traditional RAG does one search, so it probably gives you the sales numbers but misses the recalls. Agentic RAG searches sales first, then recalls, then combines both into one answer.

The rule: use traditional RAG for simple lookups. Use agentic RAG when the question needs multiple steps or multiple sources.

Assignment

- In this assignment, you will build and compare two RAG systems: a regular RAG pipeline and an agentic RAG agent. The goal is to show how a model uses your dataset, and how an agent decides when and where to search.
- Dataset
 - You must use the following dataset: Kaggle Bank Customer Complaints
- Tasks
 1. Build a regular RAG pipeline: chunk the complaints, retrieve top matches for a query, and generate an answer.
 2. Build an agentic RAG agent with at least two tools (e.g., search by category, search by date). Implement a planner and a loop that can call multiple tools before answering.
 3. Run both on the same three queries, including one that needs multiple sources.
 4. Compare the results step by step.

Summary

- ✓ Regular RAG vs Agentic RAG
- ✓ Fixed pipeline vs agent-driven retrieval
- ✓ Planning and tool use
- ✓ Multi-step search
- ✓ Comparing steps, sources, speed, and query handling
- ✓ When to use each
- ✓ Assignment: build and compare both systems

Thank You

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